MILITARY SPECIFICATION

PROJECTILE, 175 MILLIMETER,

WP, XM510E1; FILLING ASSEMBLY

1. SCOPE

1.1 This specification covers the filling of a projectile with white phosphorus (WP).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-215 - Chemical Agent, White Phosphorus (WP).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition.

DRAWINGS

US ARMY MUNITIONS COMMAND

PICATINNY ARSENAL

P8888049 - Filling Assembly.
F8857344 - Pallet for Projectile, 175MM.

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing P8888049 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the projectile shall conform to the specifications and drawings listed on Drawing P8888049 and subsidiary drawings.

3.2 Filling and assembly.

3.2.1 Assembly. The projectile shall be assembled in accordance with Drawing P8888049.

3.2.2 Filling. The projectile, clean and free of foreign matter, shall be filled with molten WP conforming to MIL-C-215 (see 6.2 and 6.3). When water is in contact with the WP during the filling operation, the water shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0. After test for leakage, the WP shall be solidified with the projectile in a vertical nose-up position.

3.2.3 Filling height. There shall be a 3 to 5 percent void, by total volume, existing above the surface of the water over the WP in the filled and closed projectile. The depth of water remaining in the filled and closed projectile shall not exceed 1/4 inch. The projectile shall be radiographed in a vertical nose-up position while the temperature of the WP is 130° ± 10° F when tested as specified in 4.2.4.1 to determine filling void and seating of the burster casing (see 6.2).

3.3 Cleaning after closing. The exterior surface of the filled and closed projectile, and the interior of the burster casing shall be cleaned thoroughly to remove all grease, oil and WP contamination (see 6.4).

3.4 Leakage. When tested as specified in 4.2.4.2 the closed and cleaned projectile shall not leak when maintained at a temperature of 160° ± 5° F for a period of 8 hours. Alternate method: The projectile shall not leak when maintained at a temperature of 207° to 212° F for a minimum period of 15 minutes.

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3.5 Workmanship. The projectile shall be free of all foreign matter (oil, grease, contamination) and damage such as dents, cracks and deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. Contained in any one interfix number of projectiles, no more than one interfix number from one manufacturer of the following components is permitted.

a. Burster casing

4.2.2. Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination and tests. The sample projectiles and the level A preparation for delivery shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 For critical defects. Each item in the lot shall be tested for the critical characteristics listed in the classification of defects.

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4.2.3.3 Classification of defects.

(a) Filling assembly (Drawing P8888049).

Categories	Defects	Acceptance standards
Critical:		
1	Leakage	4.2.4.2
2	WP contamination	
Major: AQL 1.0 percent defective		
101	Burster casing not seated	4.2.4.1
102	Threads damaged or contaminated	
103	Bourrelet diameter (maximum) after protective finish	
104	Filling excessive	4.2.4.1
105	Filling insufficient	4.2.4.1
106	Excess water	4.2.4.1
107	Marking incorrect	
108	Protective finish missing	
109	Workmanship (3.5)	
Minor: AQL 2.5 percent defective		
201	Rotating band damaged	
202	Marking missing or illegible	
203	Threads not coated	

(b) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major: AQL 4.0 percent defective	
101	Pallets incorrect
102	Strapping loose or missing
103	Marking illegible or incorrect

4.2.4 Tests.

4.2.4.1 Filling height. The projectile shall be radiographed while the WP is molten as specified and the projectile stored in a vertical nose-up position.

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4.2.4.2 Leakage. The projectile shall be placed in an oven in a vertical nose-down position (for alternate method, nose-up position) and maintained at the specified temperature for the required period of time. The temperature shall be measured by thermocouples or thermometers inserted in the burster wells of at least two projectiles located in the center of each oven bay batch. Leakage to be determined by visual examination.

5. PREPARATION FOR DELIVERY

5.1 Packing, level A. The filled and closed projectile shall be palletized in accordance with Drawing 8857344.

5.2 Marking. In addition to any special marking required by the contract or order, pallets shall be marked in accordance with Drawing 8857344.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Supplier need not furnish radiographic equipment.
(c) The Government will furnish the inspection standard.

6.2 Caution. Due to the hazardous nature of WP, filling installations are advised to take all possible precautions to protect personnel from injury while filling projectiles.

6.3 Filling aid. Projectiles filled with approximately 29 pounds of WP have been found to comply to 3.2.3. This information is furnished as an aid to procurement and not as a criterion for filling.

6.4 Decontamination. The projectiles may be decontaminated by hot water and detergent in combination with steam lances and wire brushes. This information is furnished as an aid to production and not as a filling line design criteria.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1320-A112

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